

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, adherent	Molecules Electroporated	DNA: pAG-6 (pSV2-gpt derivative containing hamster APRT), 8.4 kB, linear.
Species Used	Hamster, CHO -ATS49, ovary		

Before the Pulse

Cell Growth Medium	DMEM + 10% Fetal Bovine Serum + Non Essential Amino Acids (NEAA)+ Pen/Strep (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Log phase
		Pre-pulse Incubation	10 min., 4°C
Wash Solution	TD (analogous to Phosphate Buffered Saline)		

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature	4 °C		
Electroporation Medium*	Berg buffer (HEPES Buffered Saline- see notes)	Cuvette Gap	0.4 cm
Cell Density	Not given	Voltage	0.8 kV
Volume of Cells	10 (7) cells / 0.8 mls	Field Strength	2.0 kV/cm
DNA Concentration	1µg / µl		
DNA Resuspension Buffer	TE (10 mM Tris, 1 mM EDTA, pH 8.0)	Capacitor	25 µF
Volume of DNA	20 µl	Resistor	(Pulse Controller) Ω none
		Time Constant	0 .9 msec

After the Pulse

Outgrowth Medium	DMEM + 10% Fetal Bovine Serum + NEAA + Pen/Strep
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Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37 °C
Length of Incubation	14 days
Selection Method or Assay Used	Alanosine, Azaserine, Adenine (ap ^{rt} +) or HAT (gpt+)
Electroporation Efficiency	2 x10 (7) cells / 20 µg
Per Cent Survival	usually high %

Pennington, S.L., and Wilson, J.H. (1991) Gene targeting in Chinese Hamster Ovary cells is conservative. *PNAS* **88**: 9498-9502.

HBS: 10mM HEPES, pH 7.2,150 mM NaCl, 5 mM CaCl₂

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